

Cytotoxicity of Mouthrinses on Epithelial Cells by Micronucleus Test



Objectives To determine the cytotoxicity of three commercial mouthrinses Klorhex, Andorex and Tanflex on buccal epithelial cells using micronucleus (MN) test. **Materials and Methods** 28 patients with aged 16–24 undergone three mouthrinses' application were analyzed before and after one week exposure. Physiologic saline was used for the control group. The MN incidence was scored in the buccal epithelial of each participants. The difference in pre- and post-treatment after one week incidence of MN and plaque (PI) and gingival indices (GI) was compared by non-parametric statistical tests. **Results** The micronuclei incidence increased in Klorhex, Tanflex and Andorex groups after exposure to mouth rinses ($P < .05$). But when compared with the control group, there was not any difference between Andorex and control group ($P > .05$). In the other study groups, MN incidence was significantly increased after 7 days treatment ($P < .05$). GI scores of all groups were decreased significantly ($P < .05$). PI scores were decreased only in the Klorhex group ($P < .05$). **Conclusions** Our primary findings support the presence of possible cytotoxic effects of the mouthrinses on gingival epithelial cells.